SQL OPERATIONS ON DATASET-2

Creation of tables(customers,orders,products):

```sql
-- 1. Create Customers table
CREATE TABLE customers (
    customer_id int PRIMARY KEY,
    customer_name VARCHAR(50),
    city VARCHAR(50)
);

-- 2. Create Orders table
CREATE TABLE orders (
    order_id int PRIMARY KEY,
    customer_id int,
    order_date DATE,
    total_amount double(10,2),
    FOREIGN KEY (customer_id)
        REFERENCES customers(customer_id)
);

-- 3. Create Products table
CREATE TABLE products (
    product_id int PRIMARY KEY,
    product_name VARCHAR(100),
    price double(10,2),
    stock int
);
```

Inserting data into tables:

```
-- Insert customers
INSERT INTO customers VALUES (1, 'Rahul', 'Hyderabad');
INSERT INTO customers VALUES (2, 'Priya', 'Chennai');
INSERT INTO customers VALUES (3, 'Anudeep', 'Warangal');
INSERT INTO customers VALUES (4, 'Sneha', 'Bengaluru');
INSERT INTO customers VALUES (5, 'Kiran', 'Hyderabad');
INSERT INTO customers VALUES (6, 'Meena', 'Vijayawada');

-- Insert orders
INSERT INTO orders VALUES (101, 1, DATE '2026-10-01', 25000);
INSERT INTO orders VALUES (102, 2, DATE '2026-10-03', 12000);
INSERT INTO orders VALUES (103, 1, DATE '2026-10-05', 8000);
INSERT INTO orders VALUES (104, 3, DATE '2026-10-07', 45000);
INSERT INTO orders VALUES (105, 4, DATE '2026-10-10', 18000);
INSERT INTO orders VALUES (106, 5, DATE '2026-10-12', 6000);
INSERT INTO orders VALUES (107, 2, DATE '2026-10-15', 32000);

-- Insert products
INSERT INTO products VALUES (201, 'Laptop', 65000, 10);
INSERT INTO products VALUES (202, 'Smartphone', 25000, 20);
INSERT INTO products VALUES (203, 'Headphones', 2500, 50);
INSERT INTO products VALUES (204, 'Keyboard', 1500, 35);
INSERT INTO products VALUES (205, 'Monitor', 18000, 12);
INSERT INTO products VALUES (206, 'Mouse', 800, 60);
INSERT INTO products VALUES (207, 'Tablet', 30000, 8);

COMMIT;
```

select * from customers;

customer_id	customer_name	city
1	Rahul	Hyderabad
2	Priya	Chennai
3	Anudeep	Warangal
4	Sneha	Bengaluru
5	Kiran	Hyderabad
6	Meena	Vijayawada
NULL	NULL	NULL

select * from orders;

order_id	customer_id	order_date	total_amount
101	1	2026-10-01	25000.00
102	2	2026-10-03	12000.00
103	1	2026-10-05	8000.00
104	3	2026-10-07	45000.00
105	4	2026-10-10	18000.00
106	5	2026-10-12	6000.00
107	2	2026-10-15	32000.00
NULL	NULL	NULL	NULL

select * from products;

product_id	product_name	price	stock
201	Laptop	65000.00	10
202	Smartphone	25000.00	20
203	Headphones	2500.00	50
204	Keyboard	1500.00	35
205	Monitor	18000.00	12
206	Mouse	800.00	60
207	Tablet	30000.00	8
NULL	NULL	NULL	NULL

1)

```sql
/*Find customers who live in Hyderabad.*/
SELECT *
FROM customers
WHERE city = 'Hyderabad';
```

Output:

customer_id	customer_name	city
1	Rahul	Hyderabad
5	Kiran	Hyderabad
NULL	NULL	NULL

2)

```sql
/*Display each order along with the customer's name.*/
SELECT o.order_id,
       c.customer_name,
       o.total_amount
FROM orders o
INNER JOIN customers c
    ON o.customer_id = c.customer_id;
```

Output:

order_id	customer_name	total_amount
101	Rahul	25000.00
103	Rahul	8000.00
102	Priya	12000.00
107	Priya	32000.00
104	Anudeep	45000.00
105	Sneha	18000.00
106	Kiran	6000.00

3)

```sql
/*Calculate the total amount spent by each customer.*/
SELECT c.customer_name,
       SUM(o.total_amount) AS total_spent
FROM customers c
JOIN orders o
    ON c.customer_id = o.customer_id
GROUP BY c.customer_name;
```

Output:

customer_name	total_spent
Rahul	33000.00
Priya	44000.00
Anudeep	45000.00
Sneha	18000.00
Kiran	6000.00

4)

```sql
/*Display products from highest to lowest price.*/
SELECT product_name, price
FROM products
ORDER BY price DESC;
```

Output:

product_name	price
Laptop	65000.00
Tablet	30000.00
Smartphone	25000.00
Monitor	18000.00
Headphones	2500.00
Keyboard	1500.00
Mouse	800.00

5)

```sql
/*Find products priced above the average product price.*/
SELECT product_name, price
FROM products
WHERE price > (
    SELECT AVG(price)
    FROM products
);
```

Output:

product_name	price
Laptop	65000.00
Smartphone	25000.00
Tablet	30000.00

6)

```sql
/*Find customers who have never placed an order.*/
SELECT c.customer_id, c.customer_name
FROM customers c
LEFT JOIN orders o
    ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

Output:

customer_id	customer_name
6	Meena

7)

```sql
/*Find cities having more than one customer.*/
SELECT city,
       COUNT(*) AS total_customers
FROM customers
GROUP BY city
HAVING COUNT(*) > 1;
```

Output:

city	total_customers
Hyderabad	2